

ST. JOSEPH UNIVERSITY IN TANZANIA
ORDINARY DIPLOMA IN ENGINEERING PROGRAMMES
BASIC CONCEPTS OF COMPUTERS

Time: 2:30 Hours

Max. Marks: 100

PART - B (5 X 4 = 20 Marks)

(Answer any five)

11. Explain the data processing in the following terms:
(i) Data processing system (ii) Storing data
12. What are the differences between program and packages?
13. Differentiate between application and system software.
14. Write short notes on the following:
(i) Securities and Protection (ii) Management tools
15. Convert $(100010)_2$ to an octal number.
16. Convert $(242)_{10}$ into hexadecimal.
17. Find decimal equivalent of following binary numbers:
(i) $(111111)_2 = (?)_{10}$ (ii) $(110100)_2 = (?)_{10}$ (iii) $(111000)_2 = (?)_{10}$ (iv) $(110101)_2 = (?)_{10}$
18. Explain the role of input-output device in computer system.
19. Explain all the four types of operating systems.
20. Explain the various features of computer system.

PART - C (5 X 12 = 60 Marks)

(Answer any five)

21. Explain different types of computer generation based on processing speed with its characteristics.
22. (a) Explain any five input and five output devices.
(b) What are the differences between primary and secondary storage devices?
23. What do you mean by digital system? Explain different types of digital system.
24. Define operating system. Discuss various types of operating system.
25. Convert $(11100011)_2$ to hexadecimal.
26. Explain at least eight (8) pointing devices.
27. What is printer? Explain the types of printer.
28. Explain in detail the history of computer; also explain the evolution process of it.
29. Convert the following Number System.

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a) $(534)_8 = (?)_{16}$ b) $(101011)_2 = (?)_8$

c) $(624)_8 = (?)_2$ d) $(11101)_2 = (?)_8$

e) $(3B1)_{16} = (?)_2$ f) $(AC2)_{16} = (?)_8$

30. What is Memory? Explain types of Memory in brief.